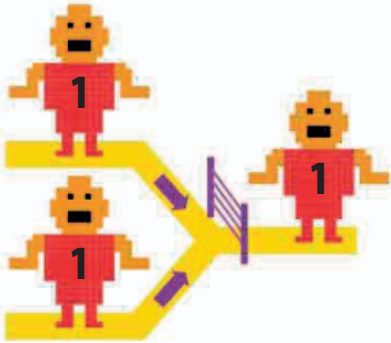


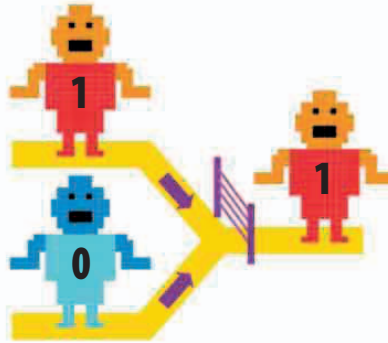
OR gate

An OR gate produces an “on” output when either one of the inputs is “on”, or when both are “on”.



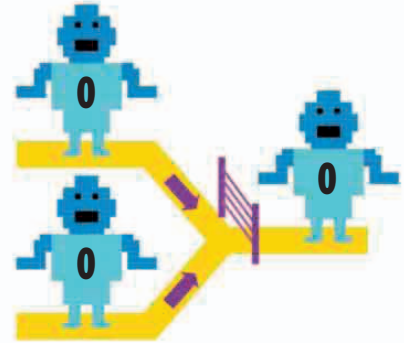
△ **Inputs 1 and 1 = output 1**

Two “on” inputs produce an “on” output.



△ **Inputs 1 and 0 = output 1**

One “on” and one “off” input still produce an “on” output.

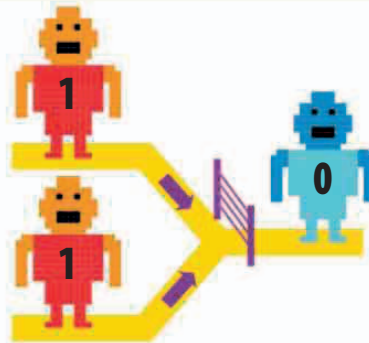


△ **Inputs 0 and 0 = output 0**

Only two “off” inputs produce an “off” output from an OR gate.

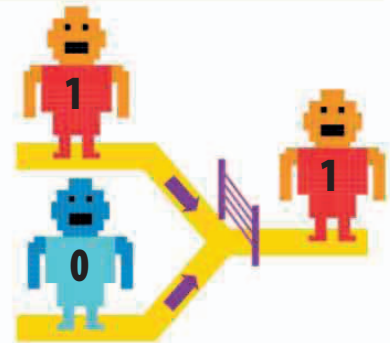
EXCLUSIVE OR gate

This type of gate only gives an “on” output when one input is “on” and the other is “off”. Two “on” or two “off” inputs will produce an “off” output. Gates like this are also known as “XOR” gates.



△ **Inputs 1 and 1 = output 0**

Two “on” inputs produce an “off” output.



△ **Inputs 1 and 0 = output 1**

The output is only “on” when the inputs are different.

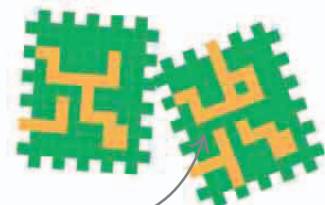


EXPERT TIPS

Building computer circuits

By combining these four basic logic gates, you can create circuits to perform a whole range of advanced functions. For example, by linking an AND gate to an XOR gate, you create a circuit that can add two binary digits (bits) together. By linking two OR gates

with two NOT gates in a loop, you can create a circuit that will store a bit of data (a single 1 or 0). Even the most powerful computers are based on billions of tiny logic circuits.



Computer chips contain many logic circuits